

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

- (1) (a) Define the following: • semi-algebra, • algebra, • σ -algebra.
 (b) Let $\mathcal{A}$ be a collection of subsets of $\Omega \neq \emptyset$. Prove that: $\mathcal{A}$ is an algebra iff it satisfies:

$$(i) \Omega \in \mathcal{A}, (ii) A, B \in \mathcal{A} \Rightarrow A \setminus B \in \mathcal{A}$$

(6×3 + 17 = 35 points)

- (2) Let $\mathcal{F}$ be a σ -algebra of subsets of $\Omega \neq \emptyset$. Take $\Omega_0 \subset \Omega$ such that $\Omega_0 \notin \mathcal{F}$. Define $\mathcal{F}_0 = \{F \cap \Omega_0 : F \in \mathcal{F}\}$. Show that $\mathcal{F}_0$ is a σ -algebra of subsets of Ω_0 .

(15 points)

- (3) Let $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$. Define $A_n = \{j : j \geq n\}$, $n \geq 1$.

(a) If $\mu(A) = |A|$ (counting measure). Show that monotone continuity from above does **not** hold for this $\{A_n\} \subset \mathcal{F}$.

(b) On the same $(\Omega, \mathcal{F})$, can you specify a *different* measure μ' such that the monotone continuity from above does hold for the *same* $\{A_n\} \subset \mathcal{F}$?

(c) In general, under what conditions does the monotone continuity from *above* hold?

(d) In general, under what conditions does the monotone continuity from *below* hold?

(4 × 10 = 40 points)

- (4) Take $F(x) = (1 - e^{-x}) \mathbf{I}_{\{x \geq 0\}}$.

(a) Using the steps in Caratheodory Extension Theorem(s), show by starting from defining a measure μ_F on a suitable semi-algebra, how one can extend it to the Lebesgue-Stieltjes measure μ_F^* on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

(b) Is such a measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ unique? Why?

[Write down the steps of the theorems carefully and state the results used in each step]

(10 + 5 = 15 points)